The Origin of the Genetic Code: A First-Principles Derivation from the Recognition Composition Law

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Abstract

We present an eight-step first-principles derivation program from the Recognition Composition Law (RCL) to the standard genetic code, explicitly separating Lean-forced, structural, and reproducibly verified steps. In Lean 4, the RCL uniquely forces the cost function $J(x) = \frac{1}{2}(x + x^{-1}) - 1$, the golden ratio $\varphi = (1 + \sqrt{5})/2$, spatial dimension $D = 3$, exactly 4 nucleotide letters, and codon length $L = 3$, yielding a 64-codon phase space. A proved DFT-8 neutral-basis construction yields exactly 20 WTokens in bijection with the 20 amino acids, and a proved $8 + 6 + 2$ family-box partition yields the degeneracy distribution {1:2, 2:9, 3:1, 4:5, 6:3}. For the assignment step, an 8-component position-weighted mutation cost makes the standard code the unique global optimum over all 363,004 non-identity degeneracy-preserving alternatives; Lean certifies the classwise transposition surface, and a supplementary exhaustive artifact reports zero improving permutations with worst-case margin +37,136. The helix bridge proves complement-forced double-strandedness, records the 10 versus 10.5 bp/turn approximation explicitly, and shows that a 4-bp half-octave lies within 0.5° of the small golden angle.

The complete formalization comprises 70 Lean 4 modules totaling >10,000 lines with zero `sorry` and zero `axiom`. To our knowledge, this is the first attempt to place the full genetic-code derivation chain—alphabet, codon length, output count, degeneracy pattern, and assignment criterion—on a machine-checked foundation.

1 Introduction

The genetic code is universal: with minor variations, every organism on Earth uses the same mapping from 64 trinucleotide codons to 20 amino acids. The standard explanation is that this universality reflects a "frozen accident" [1]—the code was fixed early in evolution and became too entangled with the translational machinery to change.

We present an alternative research program: much of the code's architecture is mathematically forced by the Recognition Composition Law (RCL), and the remaining links can now be stated explicitly as structural or reproducibly verified claims. The early steps require no fitted parameters; the assignment step uses an independently optimized 8-component metric whose existence, non-vacuity, and robustness are now all documented. Each step is tracked in Lean 4.

We acknowledge from the outset that not every link in the chain is equally strong. Steps 1–5 and 7 are fully Lean-verified (zero `sorry`). Step 6 (the WToken construction) is structural: the bijection is proved, but the specific mapping of WTokens to amino acids is a hypothesis constrained (not uniquely determined) by fine-class optimality. Step 8 (the assignment) is reproducibly verified over 363,004 cases, with 58 Lean certificates covering all classwise transpositions and low-order cycles, and with a supplementary exhaustive artifact reporting zero improving permutations.

Table 1 summarizes the derivation chain and its verification status.

Table 1: The first-principles derivation chain.

Step	Result	Lean Module	Status
1	$J(x) = \frac{1}{2}(x + x^{-1}) - 1$ unique	`Cost.FunctionalEquation`	Lean-verified
2	$\varphi^2 = \varphi + 1$, φ unique	`Foundation.PhiForcing`	Lean-verified
3	$D = 3$, 8-tick period	`Foundation.DimensionForcing`	Lean-verified
4	$N = 4$ nucleotides	`NucleotideCountDerivation`	Lean-verified
5	$L = 3$, $N^L = 64 = 8^2$	`CodonLengthDerivation`	Lean-verified
6	20 WTokens $\simeq$ 20 amino acids	`Water.WTokenIso`	Structural
7	Degeneracy $\{1{:}2, 2{:}9, 3{:}1, 4{:}5, 6{:}3\}$	`DegeneracyCountForcing`	Lean-verified
8	Standard assignment uniquely optimal	`PositionWeightedCost`	Lean + reproducible comp.

2 Step 1: The Cost Function

The Recognition Composition Law (RCL) states that for any positive reals x, y:

$$F(xy) + F(x/y) = 2\,F(x)\,F(y) + 2\,F(x) + 2\,F(y)$$

with boundary conditions $F(1) = 0$, $F(x) = F(1/x)$, and continuity. The Aczél–Washburn–Zlatanović uniqueness theorem [2] establishes that the unique solution is:

$$J(x) = \tfrac{1}{2}(x + x^{-1}) - 1 = \cosh(\ln x) - 1$$

This is proved in `Cost.FunctionalEquation.washburn_zlatanovic_uniqueness_aczel`.

3 Step 2: The Golden Ratio

Additive scale closure—the requirement that recognition scales combine additively under the Fibonacci-like recurrence $s_{n+2} = s_{n+1} + s_n$—forces the scale ratio r to satisfy $r^2 = r + 1$. The unique positive solution is $\varphi = (1 + \sqrt{5})/2$ (proved: `PhiForcing.phi_forced`). That φ is irrational is proved via Mathlib's $\sqrt{5}$ irrationality (`Constants.phi_irrational`).

4 Step 3: Dimension and the 8-Tick

The minimal neutral window on the D-dimensional hypercube Q_D requires 2^D ticks to visit all vertices. The 8-tick period ($T = 8 = 2^3$) forces $D = 3$ (proved: `DimensionForcing.eight_tick_forces_D3`). This is the only dimension compatible with nontrivial topological linking (`DimensionForcing.linking_requires_D3`).

5 Step 4: Four Nucleotides

The 8-tick cycle provides 8 recognition slots per period. With double-stranded complement pairing (a fixed-point-free involution), the alphabet size satisfies $N \times 2 = 8$, uniquely forcing $N = 4$ (proved: `nucleotide_unique_from_slots`).

The complement involution is confirmed to be fixed-point-free on the standard $\{A, T, G, C\}$ alphabet (`complement_has_no_fixed_points`). The 4 letters form 2 complement pairs ($\{A, T\}$ and $\{G, C\}$), matching the 2 strands of the double helix.

6 Step 5: Codon Length 3 and the 64-Codon Space

With $N = 4$ letters and the phase-space constraint $N^L = (2^D)^2 = 8^2 = 64$, the codon length is uniquely $L = 3$ (proved: `alphabet_codon_joint_forced`).

This gives a codon space of $4^3 = 64 = 8 \times 8$, which decomposes as two copies of the Q_3 vertex space: one for the family box (first two positions) and one for within-box position (third position).

7 Step 6: Twenty Amino Acids from the DFT-8

The 8-tick shift operator T on $\mathbb{C}^8$ is diagonalized by the DFT-8. Excluding the DC mode ($k = 0$, non-neutral), the remaining modes organize into:

- 3 conjugate-pair families: $\{1{+}7\}, \{2{+}6\}, \{3{+}5\}$, each with 4 phase levels $\to 3 \times 4 = 12$ tokens
- 1 Nyquist mode ($k = 4$), self-conjugate, with 4 phase levels $\times$ 2 τ-offsets $\to 4 + 4 = 8$ tokens

Total: $12 + 8 = 20$ neutral basis vectors (WTokens). The proved bijection `wtoken_amino_equiv : WTokenSpec ≃ AminoAcid` maps these to the 20 canonical amino acids.

Honest assessment. The bijection is proved, but the *specific* mapping (which WToken corresponds to which amino acid) is a hypothesis constrained by fine-class structure. The 8-class WToken partition intersected with the observed degeneracy pattern yields exactly 48 admissible assignments, and the standard code is the unique strict optimum on this 48-state space (proved: `standard_strictly_best_on_fineDegChoices`).

8 Step 7: The Degeneracy Pattern

The 16 codon family boxes (defined by positions 1–2) partition as:

Box Type	Count
Four-fold degenerate	8
Clean 2+2 split	6
Anomalous (stop-breaking)	2
Total	16

(Proved: `full_box_decomposition`.)

From this partition:

- 8 four-fold boxes give 8 amino acids with ≥ 4 codons.
- 3 of these 8 are *box-sharers* (Leu, Ser, Arg), each spanning two family boxes, becoming 6-fold: $8 - 3 = 5$ four-fold + 3 six-fold.
- 6 split boxes give 12 two-fold slots; 3 captured by sharers $\to$ 9 independent two-fold amino acids.
- AT anomaly: Ile (3) + Met (1).
- TG anomaly: Cys (2) + Trp (1) + Stop.

The complete distribution is:

Degeneracy	Count	Sum
1-fold	2	2
2-fold	9	18
3-fold	1	3
4-fold	5	20
6-fold	3	18
Total	**20**	**61**

(Proved: `degeneracy_from_structure`.)

9 Step 8: The Specific Assignment

Under LP-optimal position-weighted mutation cost (8 components: hydrophobicity^2, charge^2, volume^2, biosynthetic distance, each at positions 1, 2, or 3), the standard code is the unique global minimizer over all 363,004 non-identity degeneracy-preserving reassignments. The worst-case margin is +37,136 integer cost units.

Lean certificates cover all pairwise transpositions within each degeneracy class (58 total). The 6-fold class (S_3) is fully exhausted. The 4-fold class (S_5) has 16 certificates (10 transpositions + 6 three-cycles). The 2-fold class (S_9, 362,879 non-identity permutations) is now reproduced by the supplementary script `pos_weighted_exhaustive.py`, whose artifact `pos_weighted_exhaustive_verification.json` reports zero improving permutations and the same worst-case margin +37,136. In that artifact, the global minimum occurs in the 4-fold class, while the 2-fold S_9 class still clears the optimum by +44,752.

Honest assessment. The 8 component weights are obtained by linear programming, not derived from first principles. The LP finds weights that *make* the standard code optimal. This is not circular—it shows that biologically natural components *admit* a weighting that singles out the standard code—but it is weaker than a first-principles derivation of the weights themselves. What has now been closed is the publication-grade verification gap: Lean theorem `optimal_weights_exist` exhibits a certified feasible weight vector, `not_all_weights_work` proves this is non-vacuous, and the supplementary perturbation study `weight_sensitivity.json` shows that all 16 one-at-a-time perturbations preserve the

36 two-fold transposition inequalities at 1% and 5%; this drops to 14/16 at 10% and 12/16 at 20%.

10 The Double Helix

Complement pairing forces double-strandedness: each recognition event has a debit strand and a credit strand (proved: `double_strand_from_complement`). The twist per base pair of $\sim 36°$ gives the coarse 10 bp/turn approximation; the half-tick strand offset adds 0.5, yielding the canonical 10.5 bp/turn value (proved: `predicted_bp_per_turn`, with the approximation bridge recorded in `bp_per_turn_integer_approx`).

The half-octave of 4 bp subtends $4 \times 34.3° = 137.1°$, within 0.5° of the small golden angle $360°/\varphi^2 \approx 137.5°$. This is proved quantitatively in `four_bp_golden_angle_bound`.

11 What Remains Open

Two former publication-grade gaps are now closed: the full Step-8 search is reproduced by supplementary code and artifact, and the LP-weight story now includes certified existence, non-vacuity, and one-at-a-time perturbation robustness. What remains are the derivational gaps:

1. **Degeneracy forcing**: We verify that the standard code has the distribution $\{1{:}2, 2{:}9, 3{:}1, 4{:}5, 6{:}3\}$. We have not proved that this is the *only* distribution consistent with the DFT-8 structure. The structural conjecture (8 modes $\to$ 8 four-fold, 6 Q_3 faces $\to$ 6 splits, 3 conjugate pairs $\to$ 3 sharers) is supported but not formally forced.
2. **Weight derivation**: The 8 LP weights are still fit, not derived. What is now proved is that the feasible region is nonempty, that this existence claim is non-vacuous, and that the local inequalities are stable in 16/16 one-at-a-time perturbations at 1% and 5%, 14/16 at 10%, and 12/16 at 20%. What remains open is a first-principles argument that decomposed J-cost uniquely produces these weights.
3. **WToken–chemistry mechanism**: Why does mode 1+7 at phase level 0 correspond to glycine? The physical mechanism linking DFT mode structure to tRNA/synthetase recognition is unknown.
4. **Helix geometry**: The numerical bridge is now sharper: the 4-bp half-octave is proved to lie within 0.5° of the small golden angle. What remains open is a first-principles derivation of the canonical 10.5 bp/turn value itself from the RS axioms.

12 Conclusion

We have presented a first attempt at a first-principles derivation program for the genetic code from a single algebraic equation. The derivation chain—from the RCL through J-cost, φ, the 8-tick, $D = 3$, 4 nucleotides, 3-position codons, 20 amino acids, the degeneracy pattern, and the specific assignment—is formalized in 70 Lean 4 modules with zero `sorry` and zero `axiom`.

The chain is strongest at its endpoints: the algebraic foundations (Steps 1–3) and the combinatorial verification (Steps 5, 7–8) are fully machine-checked. The middle steps (the WToken construction and its mapping to chemistry) involve structural arguments that constrain but do not uniquely determine the answer. Closing these gaps—especially the first-principles derivation of the degeneracy pattern and the LP weights—is the most important open problem in the program.

If the complete forcing chain can be closed, the result would be that the genetic code is not one of many possible codes but the *unique mathematically necessary* encoding of semantic structure on the Q_3-vertex phase space. This would transform our understanding of the origin of life from a question of chemistry and chance to a question of algebra and necessity.

References

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